

Chebyshev Greedy Algorithm for Nonlinear PDEs

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Abstract. We study the Chebyshev greedy algorithm (CGA) for convex nonlinear partial differential equation energies by shallow ReLU^k ridge functions, selected from finite candidate pools with inexact coefficient solves. A one-step estimate separates four contributions: coverage of the continuous dictionary by the pool, evaluation of the selection score, radial stationarity in the active space, and the accuracy of the coefficient optimization. For the nested spaces generated by the algorithm, an approximate-source projection bound keeps both the source remainder and the observed orthogonal increments. For the one-dimensional $p = 4$ energies, an exact expansion connects this bound to nonlinear energy errors. A separate power-type decrease hypothesis gives an explicit projection bound on a finite index range. For standard p -Laplacian energies with $p \geq 2$, an energy exponent α yields natural-distance and Sobolev exponents $\alpha/2$ and α/p . Eight one- and two-dimensional problems illustrate the estimates. Five of them compare finite-pool CGA prefixes with finite elements and random features at prescribed retained coefficient counts, with FEM interpolation where necessary. The comparison batch archives every coefficient state and uses metric-specific quadrature checks. For the two quartic auxiliary models, conservative calibration-range bounds cover the computed errors on $N = 8\text{--}32$. Continuing the same prefixes to $N = 64$ with fixed constants, the normalized-innovation sufficient condition holds on 27 and 28 of 32 transitions, respectively. These numerical checks do not establish the same theorem guarantee throughout the continuation range or an unconditional asymptotic convergence rate.

Key words. Chebyshev greedy algorithm; nonlinear partial differential equation; shallow neural network; p -Laplacian; natural distance; finite dictionary; orthogonal greedy approximation

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1. Introduction. Variational formulations turn a broad class of elliptic boundary-value problems into convex energy minimization. Classical finite elements approximate the admissible space by locally supported polynomials, and the relation between polynomial degree, mesh scale, and degrees of freedom is well understood [3]. Neural and ridge-function methods instead construct global nonlinear trial spaces, so that the approximation problem concerns the selection and reoptimization of atoms rather than the design of a mesh. The deep Ritz method illustrates the reach of variational neural solvers [7]. The present work analyzes a deterministic greedy selection rule for such nonlinear elliptic energies.

Orthogonal greedy approximation (OGA) has a sharp Hilbert-space theory. Under suitable source and entropy conditions it produces optimal approximation rates [14, 9]. Chebyshev-type greedy methods extend the basic descent argument to convex energies in Banach spaces [17, 18], and biorthogonal or inexact variants show how selection and correction errors enter the energy recursion [5]. These results supply the approximation theory on which the analysis below relies. Applying them to a nonlinear partial differential equation (PDE) whose atoms are drawn from a finite candidate pool also requires accounting for numerical quadrature, finite dictionary approximation, and inexact coefficient minimization. The three effects are distinct: quadrature perturbs the evaluated score, a finite pool replaces the continuous supremum by a maximum

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over the pool, and an active-space solver returns only an approximate minimizer.

The dictionary used below consists of shallow ReLU^k ridge functions. Here $\text{ReLU}^k(s) = [s]_+^k$, with $[s]_+ = \max\{s, 0\}$, and the domain is a bounded subset of $\mathbb{R}^d$. Greedy neural training has been used successfully for PDEs [13]. A pool that is drawn once and then held fixed during the greedy solve provides a way to formulate weak selection, by which we mean selection of an atom whose score is a prescribed fraction of the best score available in the full dictionary: [19, Theorem 3.1] prove a convex-energy bound under Hilbert smoothness and source/entropy assumptions, while their Theorems 5.1 and 5.2 quantify dictionary sizes that realize weak selection. Their analysis assumes exact coefficient minimization over the active span. The finite-pool inequality below separates the correction error from the score and coverage errors. The universality of such constructions for nonlinear convex variational problems is examined by [2]. Variation spaces and sharp entropy estimates for shallow networks provide the source and width information used in the conditional estimates [15, 16, 10].

The analysis has three parts. First, a one-step descent estimate separates pool coverage, score quadrature, radial stationarity, and coefficient-correction errors. Second, projection bounds for the computed nested spaces retain the source remainder and the observed orthogonal increments; an exact quartic expansion connects these bounds to nonlinear energy errors. Third, Section 4.3 gives a local Hessian comparison for a concrete semilinear family and identifies the relative-score, invariant-neighbourhood, and entropy assumptions needed for a conditional convergence rate. The entropy estimate concerns the same projected, unrenormalized restarted dictionary throughout. The numerical results evaluate the finite-trajectory bounds and selected transfer quantities; they do not establish an unconditional asymptotic rate.

The convex Chebyshev algorithms of [17, 18] and the inexact weak-greedy framework of [5] are closest to the present setting. The former supplies the generic descent argument, and the latter shows how approximate selection and correction enter an energy recurrence. The Hilbert OGA results of [14, 9] apply after a local transfer to a fixed Hessian metric; they do not by themselves control a nonlinear PDE score or selection from a finite pool. [19] quantify deterministic and randomized finite dictionaries under weak selection, but they use exact active-space minimization and do not analyze a degenerate p -energy. The present work examines the additional conditions needed for nonlinear scores, finite pools, and inexact active-space minimization.

For p -growth energies the metric that matches the energy is not the ambient Sobolev norm. The nonlinear map

$$V(\xi) = |\xi|^{(p-2)/2} \xi$$

defines the natural distance, as in the finite-element analysis of the p -Laplacian [1, 6, 8]. A squared V -distance has the scale of an energy gap, so an energy bound of order $m^{-\alpha}$ gives a V -distance bound of order $m^{-\alpha/2}$ and, for $p \geq 2$, a $W^{1,p}$ -error bound of order $m^{-\alpha/p}$. This conversion has the same form for every p , although the energy exponent may depend on p . We prove the equivalence for the standard pure, regularized, and reaction p -energies and apply the corresponding conversion to each of them.

The atomic source condition of Assumption 2.2 and the directional smoothness estimate of Assumption 4.1 apply to convex energies and yield an $O(m^{-1})$ energy bound for the ideal continuous CGA. The same arguments produce an explicit one-step inequality for finite-pool CGA, in which four contributions appear separately: continuous-dictionary coverage, score quadrature, radial active-space stationarity, and coefficient optimization.

Two conditional refinements use additional residual or Hessian information. The residual-comparison estimate combines power-type convexity with a lower bound for the dictionary dual seminorm in terms of the full dual norm. A local Hessian transfer uses coercivity, a dictionary norm comparison, and a relative score condition to obtain a weak-OGA estimate in a frozen Hilbert metric. Neither set of assumptions implies the other, and both are stated as conditional estimates.

The finite trajectory produced by the algorithm can also be analyzed without requiring uniform Hessian coercivity. For a fixed finite symmetric dictionary and a fixed starting index m_0 , an explicit approximation of the source with coefficient budget B and remainder σ gives a projection estimate

$$r_{m_0+j}^2 \leq \sigma^2 + \frac{4B^2}{\mathcal{W}_j}, \quad \mathcal{W}_j = \sum_{m=m_0}^{m_0+j-1} \frac{\theta_m^2}{\ell_m^2}.$$

Here r_m is the norm of the projection error at state m , ℓ_m is the innovation norm of the selected candidate, and $0 \leq \theta_m \leq \theta_m$ records the selected direction against the best score available in the frozen dictionary. For the one-dimensional $p = 4$ energies, whose density is quartic in the gradient, an exact expansion connects this projection error to the energy gap. The estimate applies to the active spaces actually generated and carries a source remainder; it does not assume that the target lies in the finite dictionary. The stronger initial-value form of the bound is stated and evaluated numerically. A second estimate uses the normalized innovation $\gamma_m = \theta_m S_m / \ell_m$. If $\gamma_m^2 \geq c_0 r_m^{2+2/\alpha}$, inversion of the exact projection recurrence bounds $r_{m_0+j}^2$ by $(r_{m_0}^{-2/\alpha} + c_0 j / \alpha)^{-\alpha}$. The same quartic comparison then gives the corresponding energy and metric bounds. Each condition is evaluated from the retained projection drops; on a finite index range, positivity alone does not single out a unique exponent.

The numerical study evaluates the estimates together with the conditions on which they rest. Eight representative cases cover linear reaction–diffusion, cubic and hyperbolic-sine semilinear energies, and pure, regularized, and reaction p -energies. Five of them compare finite-pool CGA, a random-feature model (RFM), and finite elements (FEM) at equal numbers of independent coefficients. Random features are used in their fixed-feature sense [12]; conditioning of linearized neural trial spaces is discussed by [4] and [11]. The comparison concerns coefficient counts and uses a common evaluator. It does not attribute an unproved convergence theorem to the compared methods and does not compare wall-clock efficiency. Across these representative cases, finite-pool CGA exhibits competitive approximation accuracy at equal numbers of independent coefficients and achieves lower errors than RFM and FEM over the higher-DOF regime in several test problems, highlighting the potential benefit of adaptive greedy atom selection.

For the finite-trajectory estimates we use the one-dimensional pure and regularized $p = 4$ trajectories. The calibration range is $N = 8$ –32, and the subsequent range uses the same stored prefixes from $N = 33$ to 64 with the calibration constants fixed in advance. The reported fractions record the number of steps satisfying the stated sufficient condition on each finite range. They provide numerical evidence on the displayed trajectory; they do not verify continuous- dictionary coverage or an entropy estimate uniformly over increasing ranges. The approximate source is constructed once at the starting index from the fixed candidate and auxiliary reference dictionary. The source remainder is small relative to the starting residual but is not negligible at the terminal scale; it therefore limits the resolution of the source upper bound without giving a lower bound on the actual algorithmic error. These two computations provide

numerical support for the finite-range behavior predicted by the Hessian-transfer framework. Although the continuous-dictionary coverage and entropy assumptions are not directly verified in these experiments, the observed convergence orders and fitted innovation exponents still offer finite-range empirical evidence for the underlying theoretical mechanism.

The remainder is organized as follows. Section 2 defines the variational setting, dictionaries, and algorithms. Subsection 4.1 proves the basic descent estimates, and Section 3 develops the energy families and natural-distance conversion for $p \geq 2$. Section 4 combines residual comparison, Hessian transfer, finite-trajectory bounds, and fractional innovation. Section 5 tests these estimates on the computed trajectories; Section 6 presents equal-DOF comparisons, and Section 7 summarizes the conclusions and their scope.

2. Variational setting, dictionaries, and algorithms. Let X be a real reflexive Banach space. The abstract descent results apply to a proper convex functional $\mathcal{E} : X \rightarrow (-\infty, +\infty]$, with effective domain $\text{dom } \mathcal{E} = \{v \in X : \mathcal{E}(v) < +\infty\}$. Differentiability is assumed on an open neighborhood of the relevant iterates in the specified energy space, on which $\mathcal{E}$ is finite and continuously Fréchet differentiable. For an energy with a restricted effective domain, the additional topology and the neighborhood are stated explicitly. When an argument invokes a Hessian, its stated neighborhood is in addition contained in the twice differentiable part of the effective domain. Suppose that $\mathcal{E}$ has a unique minimizer u^* in the admissible space. The duality pairing between X^* and X is denoted by $\langle r, v \rangle$, and the energy defect is

$$\delta(u) = \mathcal{E}(u) - \mathcal{E}(u^*), \quad \delta_m = \delta(G_m).$$

The examples in Subsection 3.1 satisfy this setting after the Neumann nullspace has been removed where necessary.

2.1. Ridge dictionaries and admissibility. For an integer activation order $k \geq 1$, a raw ridge atom has the form

$$\varphi_{w,b}(x) = \text{ReLU}(w \cdot x + b)^k, \quad (w, b) \in \mathbb{R}^d \times \mathbb{R}.$$

An admissible dictionary consists of nonzero ridge functions satisfying the constraints of X , after the corresponding linear projection. We use a symmetric normalized dictionary $\mathcal{D}_X \subset X$:

$$g \in \mathcal{D}_X \implies -g \in \mathcal{D}_X, \quad \|g\|_X = 1 \quad (g \in \mathcal{D}_X).$$

When density is needed, we also require $\overline{\text{span } \mathcal{D}_X}^X = X$. We abbreviate $\mathcal{D}_X$ to $\mathcal{D}$ unless a different normalization is specified.

For pure-gradient Neumann energies the admissible space is

$$X_{p,\circ} = \left\{ v \in W^{1,p}(\Omega) : \int_{\Omega} v \, dx = 0 \right\}.$$

Each ridge is projected onto $X_{p,\circ}$ before the normalized dictionary is formed. Since an almost constant ridge may have arbitrarily small projected norm, normalization can amplify its derivatives. Hence uniform derivative bounds for the normalized dictionary require a positive lower bound on the projected norms. Zero projected atoms are excluded from the analytical dictionary; the corresponding nondegeneracy assumption is stated whenever it is needed.

180 DEFINITION 2.1 (Atomic hull and atomic norm). *For a symmetric normalized*
 181 *dictionary $\mathcal{D} \subset X$, define*

$$182 \quad K_1(\mathcal{D}) = \overline{\left\{ \sum_{j=1}^J c_j g_j : J < \infty, g_j \in \mathcal{D}, \sum_{j=1}^J |c_j| \leq 1 \right\}}^X,$$

183 *and*

$$184 \quad \|v\|_{\mathcal{A}(\mathcal{D})} = \inf\{B > 0 : v/B \in K_1(\mathcal{D})\}.$$

185 For a bounded set $K \subset X$, its n -th dyadic entropy number is

$$186 \quad \varepsilon_n(K; X) = \inf \left\{ h > 0 : \exists x_1, \dots, x_{2^{n-1}} \in X, K \subset \bigcup_{j=1}^{2^{n-1}} (x_j + hB_X) \right\}, \quad n \geq 1,$$

187 where $B_X = \{x \in X : \|x\|_X \leq 1\}$. Thus the covering balls may have centers in X .
 188 This 2^{n-1} -ball convention is used throughout the manuscript. We write $A \lesssim B$ when
 189 $A \leq CB$ with a constant independent of the iteration index; a subscript on $\lesssim$ lists
 190 the permitted parameter dependence, and $A \simeq B$ denotes the two corresponding
 191 inequalities.

192 ASSUMPTION 2.2 (Atomic source). *There is a finite $B > 0$, with $B \geq \|u^*\|_{\mathcal{A}(\mathcal{D})}$,*
 193 *such that $u^*/B \in K_1(\mathcal{D})$.*

194 This is a property of the target relative to the *theoretically normalized* dictionary. It
 195 is not interchangeable with the coefficient ℓ^1 -norm produced by a discretely normalized
 196 algorithm. In one space dimension, Proposition 2.3 gives a sufficient condition for
 197 finite atomic norm.

198 PROPOSITION 2.3 (A one-dimensional ReLU^k source class). *Let $I = (0, 1)$, $k \geq 1$,*
 199 *$1 \leq p < \infty$, and $X = W^{1,p}(I)$. Suppose that the continuous dictionary contains the*
 200 *normalized nonzero truncated powers $(x - t)_+^k / \|(\cdot - t)_+^k\|_X$, $0 \leq t < 1$, together with*
 201 *normalized fully active ridge functions spanning $\mathbb{P}_k$. If $v \in W^{k+1,1}(I)$, then v belongs*
 202 *to the atomic space and*

$$203 \quad \|v\|_{\mathcal{A}(\mathcal{D})} \leq C_{k,p} \left(\sum_{j=0}^k \left| v^{(j)}(0) \right| + \left\| v^{(k+1)} \right\|_{L^1(I)} \right),$$

204 where $C_{k,p}$ also accounts for the fixed exponent p , the interval, and the normalization of
 205 the raw atoms. It is a continuous-dictionary source constant and is not the coefficient
 206 ℓ^1 -norm returned by a discretely normalized numerical solve.

207 The proof is supplied in Supplement S4, file `A_source_proof_full.tex`, and uses
 208 the integral remainder formula. Multivariate source and entropy inputs are instead
 209 supplied by shallow-network variation-space and width theorems [15, 16, 10]; their
 210 domain and norm hypotheses must be checked separately for each application.

211 LEMMA 2.4 (Projection and renormalization). *Let $P : X \rightarrow X_0$ be bounded and*
 212 *linear. If a subdictionary $\mathcal{D}_0 \subset \mathcal{D}$ satisfies*

$$213 \quad 0 < c_0 \leq \|Pg\|_{X_0} \leq C_0 < \infty \quad (g \in \mathcal{D}_0),$$

214 *and $\tilde{\mathcal{D}}_0 = \{Pg / \|Pg\|_{X_0} : g \in \mathcal{D}_0\}$, then*

$$215 \quad PK_1(\mathcal{D}_0) \subset C_0 K_1(\tilde{\mathcal{D}}_0).$$

216 *The same computation also shows that a bounded linear map increases every entropy*
 217 *radius by at most $\|P\|$, while the inverse normalization loses at most the factor c_0^{-1} :*

$$218 \quad K_1(\tilde{\mathcal{D}}_0) \subset c_0^{-1} P K_1(\mathcal{D}_0).$$

219 *Proof sketch.* For each atomic combination, write $Pg = \|Pg\|_{X_0} (Pg/\|Pg\|_{X_0})$ and
 220 use the upper and lower bounds on $\|Pg\|_{X_0}$. Mapping a cover through P multiplies
 221 its radius by at most $\|P\|$, while renormalization costs at most c_0^{-1} . $\square$

222 **COROLLARY 2.5** (Transfer of source and entropy inputs). *In the setting of*
 223 *Lemma 2.4, suppose $u^* = Pu^\dagger$, $u^\dagger/B \in K_1(\mathcal{D}_0)$, and*

$$224 \quad \varepsilon_n(K_1(\mathcal{D}_0); X) \leq C_{\text{ent}} n^{-\beta}.$$

225 *Then*

$$226 \quad \frac{u^*}{BC_0} \in K_1(\tilde{\mathcal{D}}_0), \quad \varepsilon_n(K_1(\tilde{\mathcal{D}}_0); X_0) \leq c_0^{-1} \|P\| C_{\text{ent}} n^{-\beta}.$$

227 *Thus a ReLU source/convex-hull entropy theorem on an extended parameter domain*
 228 *passes to the projected, renormalized feasible dictionary whenever the nondegeneracy*
 229 *bounds in Lemma 2.4 hold.*

230 *Proof sketch.* Apply the two inclusions in Lemma 2.4 to the atomic representation
 231 of $u^\dagger$ and to an entropy cover of $K_1(\mathcal{D}_0)$. The source radius gains the factor C_0 , and
 232 the covering radius gains $c_0^{-1} \|P\|$. $\square$

233 **2.2. Dictionary residuals.** For $r \in X^*$, introduce the dictionary dual seminorm

$$234 \quad \|r\|_{\mathcal{D}^*} = \sup_{g \in \mathcal{D}_X} |\langle r, g \rangle|.$$

235 This quantity measures the derivative in the available dictionary directions. It is
 236 a seminorm; if the dictionary has dense linear span, its vanishing implies that the
 237 derivative is zero.

238 **2.3. Ideal and finite-pool Chebyshev greedy algorithms.** The ideal CGA
 239 combines weak selection with exact minimization over the enlarged space.

Algorithm 2.1 Continuous weak Chebyshev greedy algorithm

Assume $\mathcal{E}(0) < \infty$, choose weakness numbers $t_m \in (0, 1]$, and set $G_0 = 0$, $\Phi_0 = \{0\}$.
 Given G_m and $\Phi_m = \text{span}\{g_1, \dots, g_m\}$:

- (i) if $\|\mathcal{E}'(G_m)\|_{\mathcal{D}^*} = 0$, stop; otherwise choose $g_{m+1} \in \mathcal{D}_X$ with descent sign such
 that

$$-\langle \mathcal{E}'(G_m), g_{m+1} \rangle \geq t_m \|\mathcal{E}'(G_m)\|_{\mathcal{D}^*};$$

- (ii) set $\Phi_{m+1} = \text{span}(\Phi_m, g_{m+1})$ and compute

$$G_{m+1} = \arg \min_{v \in \Phi_{m+1}} \mathcal{E}(v).$$

240 For a general continuous dictionary the supremum need not be attained. Thus
 241 $t_m < 1$ is understood as approximate-supremum selection. The endpoint $t_m = 1$ is
 242 used only when attainment has been proved, for example on a compact nondegenerate
 243 parameter set with continuous score. The finite-pool method stops, with a recorded

reason, if the unused pool is empty, every remaining score is below the stationarity threshold, every candidate fails the rank or acceptance test, or the attempt budget is reached. In an unconstrained active-space solve, changing an atom's sign leaves the span unchanged; the descent sign required by the proof is therefore not an additional algorithmic variable.

Exact reoptimization yields the variational orthogonality

$$(2.1) \quad \langle \mathcal{E}'(G_m), v \rangle = 0 \quad (v \in \Phi_m).$$

The second algorithm describes the computed object. It uses a finite pool $\mathcal{P}_m$, an evaluated score $\tilde{s}_m$, and a correction computed to a finite tolerance.

Algorithm 2.2 Finite-pool inexact CGA

Set $\hat{G}_0 = 0$, $\hat{\Phi}_0 = \{0\}$. At step m , form a finite admissible pool $\mathcal{P}_m \subset \mathcal{D}$. Let

$$s_m(g) = \left| \langle \mathcal{E}'(\hat{G}_m), g \rangle \right|, \quad \tilde{s}_m(g) \geq 0.$$

Choose $t_m^{\text{pool}} \in (0, 1]$ and $\hat{g}_{m+1} \in \mathcal{P}_m$ satisfying

$$\tilde{s}_m(\hat{g}_{m+1}) \geq t_m^{\text{pool}} \max_{g \in \mathcal{P}_m} \tilde{s}_m(g),$$

give it the true descent sign, and define $\hat{\Phi}_{m+1} = \text{span}(\hat{\Phi}_m, \hat{g}_{m+1})$. Compute a coefficient vector whose state obeys

$$\mathcal{E}(\hat{G}_{m+1}) \leq \inf_{v \in \hat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\text{opt}}, \quad \varepsilon_m^{\text{opt}} \geq 0.$$

2.4. Finite-pool error quantities. The continuous-dictionary supremum, the exact maximum over the finite pool, and the numerically evaluated maximum are distinct quantities. We impose the following score-error and pool-coverage conditions on any step for which a positive coverage bound is used; finiteness of the pool alone does not imply a uniform positive lower bound.

$$(2.2) \quad \sup_{g \in \mathcal{P}_m} |\tilde{s}_m(g) - s_m(g)| \leq \xi_m,$$

$$(2.3) \quad \max_{g \in \mathcal{P}_m} s_m(g) \geq \tau_m \sup_{g \in \mathcal{D}} s_m(g), \quad 0 < \tau_m \leq 1.$$

LEMMA 2.6 (True score selected from an evaluated pool). *Under (2.2) and (2.3),*

$$(2.4) \quad s_m(\hat{g}_{m+1}) \geq t_m^{\text{pool}} \tau_m \left\| \mathcal{E}'(\hat{G}_m) \right\|_{\mathcal{D}^*} - (1 + t_m^{\text{pool}}) \xi_m.$$

Proof sketch. The score-error bound is inserted before and after the approximate maximization, and the pool-coverage inequality replaces the finite-pool maximum by the continuous dictionary supremum. $\square$

If the computed correction minimizes a discrete training objective $\mathcal{E}_h$, a single terminal comparison between $\mathcal{E}_h$ and the analysis energy $\mathcal{E}$ is not a uniform error bound. The appropriate deterministic statement is the following.

Let $G_{m+1}^\#$ denote an exact minimizer of $\mathcal{E}$ over $\widehat{\Phi}_{m+1}$. For an energy level Λ_m^E chosen so that both $G_{m+1}^\#$ and the computed state $\widehat{G}_{m+1}$ are included, define the relevant sublevel set and its active-space restriction by

$$(2.5) \quad \mathcal{L}_m = \{v \in X : \mathcal{E}(v) \leq \Lambda_m^E\}, \quad \mathcal{L}_m^{\text{act}} = \widehat{\Phi}_{m+1} \cap \mathcal{L}_m.$$

Thus $\mathcal{L}_m^{\text{act}}$ is the part of the current finite-dimensional trial space on which the training and analysis energies must be compared; in particular, $\inf_{v \in \mathcal{L}_m^{\text{act}}} \mathcal{E}(v) = \mathcal{E}(G_{m+1}^\#)$.

LEMMA 2.7 (Transfer from the discrete energy to the analysis energy). *Suppose on the active-space sublevel set $\mathcal{L}_m^{\text{act}}$ in (2.5) that*

$$\sup_{v \in \mathcal{L}_m^{\text{act}}} |\mathcal{E}_h(v) - \mathcal{E}(v)| \leq \zeta_m^E.$$

If the computed state satisfies

$$\mathcal{E}_h(\widehat{G}_{m+1}) \leq \inf_{v \in \mathcal{L}_m^{\text{act}}} \mathcal{E}_h(v) + \varepsilon_m^{\text{opt,h}},$$

then Algorithm 2.2 holds with $\varepsilon_m^{\text{opt}} \leq \varepsilon_m^{\text{opt,h}} + 2\zeta_m^E$.

Proof sketch. Compare the computed state with the exact active minimizer in the common sublevel set. The uniform energy discrepancy costs at most ζ_m^E at each endpoint, so the two discrepancies add to $2\zeta_m^E$. $\square$

3. Natural geometry for variational energies. Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz. We consider diffusion and reaction equations with natural Neumann boundary conditions. The regularization parameter $\varepsilon \geq 0$ is fixed; the regularized model uses $\varepsilon > 0$. The linear and semilinear equations are

$$-\Delta u + u = f, \quad -\Delta u + u^3 = f, \quad -\Delta u + \sinh u = f.$$

The standard p -structure models are

$$-\operatorname{div} A_p(\nabla u) = f, \quad -\operatorname{div} A_{p,\varepsilon}(\nabla u) = f, \quad -\operatorname{div} A_p(\nabla u) + |u|^{p-2}u = f,$$

where $A_p(\xi) = |\xi|^{p-2}\xi$, $A_p(0) = 0$, and $A_{p,\varepsilon}(\xi) = (\varepsilon^2 + |\xi|^2)^{(p-2)/2}\xi$. The boundary flux is zero. For the pure and regularized Neumann problems, the load annihilates constants and the zero-mean representative is used. The equations are understood weakly when the load is not a regular function.

3.1. Variational energies and feasible spaces. The three p -energies are defined in (3.4)–(3.6). The remaining models have the following energies, written for $d \leq 2$:

$$(3.1) \quad \mathcal{E}_{\text{lin}}(u) = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{2} u^2 \right) dx - \langle f, u \rangle,$$

$$(3.2) \quad \mathcal{E}_{\text{cub}}(u) = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right) dx - \langle f, u \rangle,$$

$$(3.3) \quad \mathcal{E}_{\text{sinh}}(u) = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \cosh u - 1 \right) dx - \langle f, u \rangle.$$

The pairing is the energy-space duality; for a regular load it is the corresponding integral. The additive -1 in (3.3) normalizes the potential and does not change the Euler–Lagrange equation.

ASSUMPTION 3.1 (Data and feasible spaces for the six models). *The following model-specific conditions are imposed.*

- (i) *For the linear and cubic models, $f \in (H^1(\Omega))^*$. For the sinh model, the convex integral functional has a load $f \in (H^1(\Omega))^*$, and the integral functional is allowed to take the value $+\infty$, its effective domain is nonempty, and local derivative/Hessian statements are restricted to an L^∞ -bounded neighborhood.*
- (ii) *For the pure and regularized p -energies, $f \in X_{p,o}^*$, or equivalently the Neumann load annihilates constants before restriction to $X_{p,o}$.*
- (iii) *For the reaction p -energy, $f \in (W^{1,p})^*$. In the regularized model, the parameter ε in (3.5) is fixed and positive.*

PROPOSITION 3.2 (Well-posedness). *Under Assumption 3.1, (3.1)–(3.3) have unique minimizers in $H^1(\Omega)$; (3.4) and (3.5) have unique minimizers in $X_{p,o}$; and (3.6) has a unique minimizer in $W^{1,p}(\Omega)$.*

Proof sketch. The quadratic, strictly convex, and p -growth terms give coercivity on the stated feasible spaces; strict convexity gives uniqueness. The pure Neumann nullspace is removed by the zero-mean restriction. $\square$

The local Hessian structure of each model is needed for the transfer estimates of Subsection 4.3. The bilinear forms at the exact solution are

$$a_{\text{lin}}(v, w) = \int_{\Omega} (\nabla v \cdot \nabla w + vw) \, dx,$$

$$a_{\text{cub},*}(v, w) = \int_{\Omega} (\nabla v \cdot \nabla w + 3(u^*)^2 vw) \, dx,$$

$$a_{\text{sinh},*}(v, w) = \int_{\Omega} (\nabla v \cdot \nabla w + \cosh(u^*) vw) \, dx.$$

For a p -diffusion density, the Hessian matrix in gradient variables is

$$|\xi|^{p-2} I + (p-2) |\xi|^{p-4} \xi \otimes \xi$$

at nonzero ξ in the pure case, with $|\xi|^2$ replaced by $\varepsilon^2 + |\xi|^2$ in the regularized case. The local Hessian statements that follow concern $p \geq 2$.

PROPOSITION 3.3 (Sufficient continuous local Hessian conditions). *In dimensions $d \leq 2$, the cubic Hessian is locally Lipschitz as a bilinear form on H^1 , and it is coercive at any solution $u^* \neq 0$. The sinh Hessian is coercive and is locally Lipschitz on an L^∞ -bounded neighborhood, with the state perturbation measured in H^1 . For fixed $\varepsilon > 0$, the regularized p -Hessian is uniformly elliptic, with lower bound proportional to ε^{p-2} ; its local Lipschitz continuity is valid in the bounded-gradient topology $W^{1,\infty}$, not in the H^1 topology asserted for the cubic model.*

Proof sketch. In dimensions $d \leq 2$, Sobolev embedding controls the cubic coefficient on bounded H^1 neighborhoods, while an L^∞ bound controls the sinh coefficient. Differentiating the regularized radial flux gives uniform ellipticity and bounded derivative on bounded-gradient sets. $\square$

PROPOSITION 3.4 (Restricted stability for the regularized p -energy). *Let $Y_M \subset W^{1,\infty}(\Omega)$ be finite dimensional and let $\|\cdot\|_{0,M}$ be any norm on Y_M . For fixed $\varepsilon > 0$, bounded subsets of Y_M have finite restricted constants L_M on the corresponding finite-dimensional segments. If the reference norm is chosen so that the gradient seminorm is a norm on the feasible space, then $\mu_M > 0$. The resulting constants may depend on M , the inverse inequality on Y_M , and ε .*

Proof sketch. On a bounded subset of the finite-dimensional space, norm equivalence and compactness give a finite Hessian-Lipschitz constant. The positive ellipticity of the regularized density then yields a positive restricted coercivity constant in any reference norm that is a norm on the feasible space. $\square$

The frozen Hessian is constant for the linear model, locally stable for the cubic and sinh models under the stated neighborhood restrictions, uniformly elliptic for fixed $\varepsilon > 0$ on bounded-gradient sets, and potentially degenerate for the pure and reaction p -models. The transfer result therefore uses either the local Hessian hypotheses or the global p -geometry, according to the model.

Energy coercivity and frozen-Hessian coercivity are distinct. A positive reaction term does not imply that the frozen p -Hessian is uniformly positive at points where both the solution and its gradient vanish, and the continuous degeneracy of the pure model does not preclude the restricted coercivity in the corresponding finite-dimensional metric.

3.2. Standard p -structure and natural distance. For standard p -structure energies, we relate the CGA energy error to the natural distance associated with the diffusion operator. The argument uses the pointwise structure of the density in addition to its growth. The pointwise inequalities are standard; the step taken here is to integrate them into an energy-to-metric estimate for the three energy families below, together with the matching $W^{1,p}$ conversion. Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz, and let $p \geq 2$. On the zero-mean space $X_{p,\circ}$, define the pure and regularized energies

$$(3.4) \quad \mathcal{E}_p(u) = \int_{\Omega} \frac{1}{p} |\nabla u|^p \, dx - \langle f, u \rangle,$$

$$(3.5) \quad \mathcal{E}_{p,\varepsilon}(u) = \int_{\Omega} \frac{(\varepsilon^2 + |\nabla u|^2)^{p/2} - \varepsilon^p}{p} \, dx - \langle f, u \rangle, \quad \varepsilon > 0.$$

On $W^{1,p}(\Omega)$, define the reaction energy

$$(3.6) \quad \mathcal{E}_{p,R}(u) = \int_{\Omega} \left(\frac{1}{p} |\nabla u|^p + \frac{1}{p} |u|^p \right) \, dx - \langle f, u \rangle.$$

For a differentiable convex functional F , its Bregman divergence is

$$(3.7) \quad D_F(u, v) = F(u) - F(v) - \langle F'(v), u - v \rangle \geq 0.$$

It measures the part of the energy increment not accounted for by the tangent functional at v . In general $D_F(u, v) \neq D_F(v, u)$, so it is not a metric. The same notation is used pointwise for a convex density: $D_{\phi}(a, b) = \phi(a) - \phi(b) - \nabla \phi(b) \cdot (a - b)$.

For scalar arguments set $V(s) = |s|^{(p-2)/2} s$, and for vector arguments set

$$(3.8) \quad V(\xi) = |\xi|^{(p-2)/2} \xi, \quad V_{\varepsilon}(\xi) = (\varepsilon^2 + |\xi|^2)^{(p-2)/4} \xi.$$

The corresponding natural distances are

$$\begin{aligned} d_V(u, v)^2 &= \|V(\nabla u) - V(\nabla v)\|_{L^2}^2, \\ d_{V,\varepsilon}(u, v)^2 &= \|V_{\varepsilon}(\nabla u) - V_{\varepsilon}(\nabla v)\|_{L^2}^2, \\ d_{V,R}(u, v)^2 &= d_V(u, v)^2 + \|V(u) - V(v)\|_{L^2}^2. \end{aligned}$$

We use $\mathcal{E}_\bullet$ and $d_{V,\bullet}$ only as paired abbreviations for one of $(\mathcal{E}_p, d_V)$, $(\mathcal{E}_{p,\varepsilon}, d_{V,\varepsilon})$, or $(\mathcal{E}_{p,R}, d_{V,R})$. The scalar map applies to the reaction term and the vector map to gradients. The map V turns the weighted quadratic structure of a p -operator into an L^2 -type distance [1, 6, 8].

Throughout this section, $A \lesssim_p B$ means $A \leq C(p)B$, and $A \simeq_p B$ means that both inequalities hold with positive constants depending only on p ; a further subscript lists any additional permitted dependence.

LEMMA 3.5 (Pointwise p -structure). *For $p \geq 2$, constants depending only on p satisfy, for every $a, b \in \mathbb{R}^d$,*

$$(3.8) \quad \frac{|a|^p}{p} - \frac{|b|^p}{p} - |b|^{p-2} b \cdot (a - b) \simeq_p (|a| + |b|)^{p-2} |a - b|^2 \simeq_p |V(a) - V(b)|^2,$$

$$(3.9) \quad |a - b|^p \lesssim_p |V(a) - V(b)|^2.$$

The analogous statements hold for the regularized density and V_ε , with $\varepsilon > 0$ fixed.

Proof sketch. These are the standard monotonicity and radial-growth inequalities for $\xi \mapsto |\xi|^{p-2}\xi$; the regularized statements follow from the same calculation with $\varepsilon^2 + |\xi|^2$. $\square$

THEOREM 3.6 (Energy–natural-distance equivalence). *For each of the three energy–space pairs*

$$(\mathcal{E}_p, X_{p,\circ}), \quad (\mathcal{E}_{p,\varepsilon}, X_{p,\circ}), \quad (\mathcal{E}_{p,R}, W^{1,p}(\Omega)),$$

let u^* be the minimizer. Then, for every admissible u , respectively,

$$(3.10) \quad \mathcal{E}_p(u) - \mathcal{E}_p(u^*) \simeq_p d_V(u, u^*)^2,$$

$$(3.11) \quad \mathcal{E}_{p,\varepsilon}(u) - \mathcal{E}_{p,\varepsilon}(u^*) \simeq_{p,\varepsilon} d_{V,\varepsilon}(u, u^*)^2,$$

$$(3.12) \quad \mathcal{E}_{p,R}(u) - \mathcal{E}_{p,R}(u^*) \simeq_p d_{V,R}(u, u^*)^2.$$

In the corresponding spaces,

$$(3.13) \quad \|u - u^*\|_{W^{1,p}}^p \lesssim_{p,\Omega,\varepsilon} d_{V,\bullet}(u, u^*)^2,$$

where the constants in the pure and reaction cases are independent of ε .

Proof sketch. Integrate the pointwise Bregman equivalences for the gradient and reaction terms. The weak optimality of u^* cancels the load term, and Poincaré’s inequality controls the zero-mean representative in the pure and regularized cases. $\square$

COROLLARY 3.7 (Metric exponents). *Fix any one of the three energy–space–distance triples in Theorem 3.6. If a sequence G_m in that feasible space satisfies*

$$(3.14) \quad \mathcal{E}_\bullet(G_m) - \mathcal{E}_\bullet(u^*) = O(m^{-\alpha})$$

for some $\alpha > 0$, then

$$(3.15) \quad d_{V,\bullet}(G_m, u^*) = O(m^{-\alpha/2}), \quad \|G_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}).$$

Proof sketch. Apply the two-sided energy–distance equivalence and the $W^{1,p}$ control in Theorem 3.6 to the assumed energy bound, then take the corresponding square or p -th root. $\square$

Remark 3.8 (Scope). The corollary is restricted to the three standard p -Laplacian-type energies just defined. Linear, cubic, and hyperbolic-sine energies have their own quadratic or local-Hessian geometry and are not instances of this statement merely because they are convex. Conversely, an unspecified energy with p -growth need not satisfy the Bregman– V equivalence in Lemma 3.5.

4. Conditional convergence and finite-trajectory projection bounds. The analysis separates finite-pool descent from approximation in a frozen Hilbert metric. Pool coverage, score evaluation, radial stationarity, and coefficient correction enter the first part; nested-space projection and local Hessian information enter the second. The connection is conditional, and the quartic models additionally admit an exact energy identity.

Throughout, m counts accepted independent directions, m_0 is a starting index, $\widehat{G}_m$ is a computed state, $G_m^\#$ an exact active-space minimizer, and z_m a frozen orthogonal projection. All derivatives act on the specified feasible linear space.

4.1. Finite-pool descent and residual comparison. Let X be a real Banach space, $\mathcal{E} : X \rightarrow \mathbb{R}$ convex and differentiable, and u^* a minimizer. For a symmetric unit dictionary $\mathcal{D}_X$, set

$$\widehat{\delta}_m = \mathcal{E}(\widehat{G}_m) - \mathcal{E}(u^*), \quad S_m^X = \sup_{g \in \mathcal{D}_X} |\langle \mathcal{E}'(\widehat{G}_m), g \rangle|.$$

For a finite pool $\mathcal{P}_m \subset \mathcal{D}_X$ and $0 < t_m^{\text{pool}} \leq 1$, write

$$|\widetilde{s}_m(g) - s_m(g)| \leq \xi_m, \quad \widetilde{s}_m(g_{m+1}) \geq t_m^{\text{pool}} \max_{g \in \mathcal{P}_m} \widetilde{s}_m(g),$$

where $s_m(g) = |\langle \mathcal{E}'(\widehat{G}_m), g \rangle|$. Its continuous coverage is $\tau_m = \max_{\mathcal{P}_m} s_m / S_m^X$ when $S_m^X > 0$, and we set $\tau_m = 1$ when $S_m^X = 0$.

ASSUMPTION 4.1 (Directional quadratic upper bound). *For some $L > 0$, every m , every $g \in \mathcal{D}_X$, and $|\lambda| \leq 1$,*

$$(4.1) \quad \mathcal{E}(\widehat{G}_m + \lambda g) \leq \mathcal{E}(\widehat{G}_m) + \lambda \langle \mathcal{E}'(\widehat{G}_m), g \rangle + \frac{L}{2} \lambda^2,$$

Assume that the expanded active space contains the signed step used below and that the inexact active correction satisfies

$$\mathcal{E}(\widehat{G}_{m+1}) \leq \inf_{v \in \widehat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\text{opt}}, \quad \varepsilon_m^{\text{opt}} \geq 0.$$

Define $H_L(a) = \max_{0 \leq \lambda \leq 1} (\lambda a - L\lambda^2/2)$.

The radial active-space stationarity defect is denoted by

$$(4.2) \quad \left| \langle \mathcal{E}'(\widehat{G}_m), \widehat{G}_m \rangle \right| \leq \eta_m^{\text{stat}}.$$

This is the radial component of the active-space stationarity residual; the full active-space dual KKT residual is a separate quantity.

PROPOSITION 4.2 (Finite-pool descent). *Under these conditions,*

$$(4.3) \quad \widehat{\delta}_{m+1} \leq \widehat{\delta}_m - H_L(a_m) + \varepsilon_m^{\text{opt}}, \quad a_m = [t_m^{\text{pool}} \tau_m S_m^X - (1 + t_m^{\text{pool}}) \xi_m]_+.$$

Proof. The score error is inserted before and after the finite-pool maximization. Testing the enlarged active space with a signed step of length $\min\{1, a_m/L\}$ gives the displayed decrease, and the correction error is added once. $\square$

If $B_{\text{src}} > 0$, u^*/B_{src} belongs to the closed absolutely convex hull of $\mathcal{D}_X$, and the radial defect satisfies (4.2), convexity gives

$$(4.4) \quad \hat{\delta}_m \leq \eta_m^{\text{stat}} + B_{\text{src}} S_m^X.$$

Thus radial stationarity is distinct from score error and correction error.

LEMMA 4.3 (Nonautonomous scalar recurrence). *If $x_m \geq 0$ and $x_{m+1} \leq x_m - b_m x_m^q$ with $q > 1$ and $b_m \geq 0$, then, for $x_0 > 0$,*

$$(4.5) \quad x_n \leq \left[x_0^{1-q} + (q-1) \sum_{m=0}^{n-1} b_m \right]^{-1/(q-1)}.$$

If $x_0 = 0$, the sequence vanishes.

Proof. Apply convexity to $x \mapsto x^{1-q}$ while the sequence is positive and sum the resulting increments; once a term vanishes, nonnegativity gives the remaining cases. $\square$

The atomic baseline follows by assuming, for $m \geq m_0$,

$$t_m^{\text{pool}} \tau_m \geq \underline{t} > 0, \quad \eta_m^{\text{stat}} + \frac{B_{\text{src}}(1 + t_m^{\text{pool}})}{t_m^{\text{pool}} \tau_m} \xi_m \leq \vartheta \hat{\delta}_m, \quad \varepsilon_m^{\text{opt}} \leq \zeta H_L(a_m),$$

with $0 \leq \vartheta, \zeta < 1$. Equation (4.4) gives $a_m \geq \underline{t}(1-\vartheta)\hat{\delta}_m/B_{\text{src}}$. Descent is monotone and $H_L(a) \geq a^2/(2 \max\{L, A\})$ for $0 \leq a \leq A$; applying this to the displayed lower bound for a_m gives $\hat{\delta}_{m+1} \leq \hat{\delta}_m - c\hat{\delta}_m^2$ with $c > 0$. Lemma 4.3 yields $\hat{\delta}_m = O((m - m_0 + 1)^{-1})$ on the tail. The ideal continuous method is the case $\tau_m = 1$ and $\xi_m = \eta_m^{\text{stat}} = \varepsilon_m^{\text{opt}} = 0$, with selection weakness bounded below.

THEOREM 4.4 (Conditional residual-comparison rate). *Let $p > 2$, $c_U > 0$, and $\kappa_0 > 0$. Suppose $\hat{\delta}_m \geq c_U \|\hat{G}_m - u^*\|_X^p$ and $S_m^X \geq \kappa_0(m+1)^{-\chi} \|\mathcal{E}'(\hat{G}_m)\|_{X^*}$ with $0 \leq \chi < 1/2$. Assume, uniformly in m , that*

$$t_m^{\text{pool}} \tau_m \geq \underline{t} > 0, \quad (1 + t_m^{\text{pool}}) \xi_m \leq \vartheta t_m^{\text{pool}} \tau_m S_m^X, \quad 0 \leq \vartheta < 1,$$

and that the correction error satisfies $\varepsilon_m^{\text{opt}} \leq \zeta H_L(a_m)$ with $0 \leq \zeta < 1$. Together with the directional bound above, these conditions give

$$(4.6) \quad \hat{\delta}_m = O((m+1)^{-p(1-2\chi)/(p-2)}).$$

Proof. Convexity and the lower bound on $\hat{\delta}_m$ give $\|\mathcal{E}'(\hat{G}_m)\|_{X^*} \geq c_U^{1/p} \hat{\delta}_m^{(p-1)/p}$. The pool and error assumptions therefore give a selected score bounded below by a fixed multiple of $(m+1)^{-\chi} \hat{\delta}_m^{(p-1)/p}$. Descent and absorption give monotonicity. The lower score bound is therefore bounded above by a fixed A , so the same quadratic lower bound for H_L gives Lemma 4.3 with $q = 2(p-1)/p$ and $b_m \asymp (m+1)^{-2\chi}$. The sum grows as $m^{1-2\chi}$. $\square$

4.2. Energy and natural distance. For the standard pure, regularized, and reaction p -energies, Theorem 3.6 and Corollary 3.7 gives energy–distance equivalence and the $W^{1,p}$ conversion. Hence an energy estimate $\hat{\delta}_m = O(m^{-\alpha})$ implies

$$d_{V,\bullet}(\hat{G}_m, u^*) = O(m^{-\alpha/2}), \quad \|\hat{G}_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}).$$

The statement is restricted to the three specified energy densities.

4.3. Local Hessian transfer and the same-dictionary requirement. The local transfer result is the main conditional refinement. Let H be a Hilbert space on which $\mathcal{E}$ is twice Fréchet differentiable in an open convex neighborhood of its minimizer u^* . Let $A_* = \mathcal{E}''(u^*)$ be bounded, symmetric, and coercive, and set $a_*(v, w) = \langle A_* v, w \rangle$, $\|v\|_*^2 = a_*(v, v)$. Assume the closed ball $\|u - u^*\|_* \leq R_*$ lies in this neighborhood and

$$(4.7) \quad |\langle [\mathcal{E}''(u) - A_*]v, w \rangle| \leq L_* \|u - u^*\|_* \|v\|_* \|w\|_*, \quad L_* R_* \leq \frac{1}{2}.$$

For a finite-dimensional space Φ , let z_Φ be the frozen A_* -projection and $G_\Phi^\sharp$ the exact energy minimizer, both inside this ball.

LEMMA 4.5 (Taylor and projection estimates). *Under (4.7), for u in this ball,*

$$(4.8) \quad |\mathcal{E}(u) - \mathcal{E}(u^*) - \frac{1}{2} \|u - u^*\|_*^2| \leq \frac{L_*}{6} \|u - u^*\|_*^3,$$

$$(4.9) \quad \|\mathcal{E}'(u) - A_*(u - u^*)\|_{**} \leq \frac{L_*}{2} \|u - u^*\|_*^2,$$

$$(4.10) \quad \|G_\Phi^\sharp - z_\Phi\|_* \leq L_* \|z_\Phi - u^*\|_*^2.$$

Proof. Taylor's formula gives the first two inequalities. The third follows by testing the active-space first-order condition with $G_\Phi^\sharp - z_\Phi$ and absorbing the term $L_* R_* \|G_\Phi^\sharp - z_\Phi\|_*^2$. $\square$

Assume the algorithmic dictionary $\mathcal{D}_X \subset H$ satisfies

$$(4.11) \quad 0 < c_* \leq \|g\|_* \leq C_* < \infty \quad (g \in \mathcal{D}_X), \quad \mathcal{D}_* = \{g/\|g\|_* : g \in \mathcal{D}_X\}.$$

Let $\Phi_{m+1} = \Phi_m + \text{span}\{g_{m+1}\}$, where $g_{m+1} \in \mathcal{D}_X \setminus \Phi_m$, and let P_m be the a_* -orthogonal projection onto Φ_m . Set $z_m = P_m u^*$ and define $S_m^* = \sup_{d \in \mathcal{D}_*} |\langle A_*(z_m - u^*), d \rangle|$. At the exact active-space minimizer $G_m^\sharp$ define the score defect

$$\eta_m^H = \sup_{g \in \mathcal{D}_X} |\langle \mathcal{E}'(G_m^\sharp), g \rangle - a_*(z_m - u^*, g)|.$$

Lemma 4.5 yields $\eta_m^H \leq C_T \|z_m - u^*\|_*^2$ on the stated ball, with C_T uniform over the specified dictionary.

PROPOSITION 4.6 (Relative score transfer). *If nonlinear selection at $G_m^\sharp$ over $\mathcal{D}_X$ has strength $t \in (0, 1]$ and $0 < t_0 < tc_*/C_*$, then*

$$(4.12) \quad (1 + t)\eta_m^H \leq (tc_* - t_0 C_*) S_m^*$$

implies that the selected direction satisfies

$$|a_*(z_m - u^*, g_{m+1}/\|g_{m+1}\|_*)| \geq t_0 S_m^*.$$

Proof. The selected unnormalized frozen score is bounded below by $tc_* S_m^* - (1 + t)\eta_m^H$; division by the selected frozen norm costs at most C_* . This proves the claim. $\square$

For a concrete semilinear family, take $H = H^1(0, 1)$ and

$$\mathcal{E}_\lambda(u) = \int_0^1 \left(\frac{1}{2} |u'|^2 + \frac{1}{2} u^2 + \frac{\lambda}{4} u^4 \right) dx - \ell(u), \quad \lambda \geq 0,$$

For $k \geq 2$ and $0 < a < b < 1$, use the dictionary $\mathcal{D}_X = \{\pm g_t : t \in [a, b]\}$, where $g_t = (x - t)_+^k / \|(\cdot - t)_+^k\|_{H^1}$. Choose a finite signed measure μ with total variation

at most $B_{\text{src}} > 0$, set $u^* = \int_a^b g_t d\mu(t)$, and define the variational load by $\ell(v) = \int_0^1 (u^* v' + u^* v + \lambda(u^*)^3 v) dx$. This construction makes u^* the unique minimizer; it does not impose homogeneous Neumann data. The parameter restriction makes the dictionary compact. Let C_{emb} satisfy $\|v\|_\infty \leq C_{\text{emb}} \|v\|_{H^1}$ and put $M_* = \|u^*\|_\infty$. Directly bounding $3\lambda(u^2 - (u^*)^2)vw$ gives

$$L_\lambda = 3\lambda C_{\text{emb}}(C_{\text{emb}}R_* + 2M_*), \quad c_* = 1, \quad C_\lambda = (1 + 3\lambda M_*^2)^{1/2}, \quad B_\lambda = C_\lambda B_{\text{src}}.$$

Thus $u^* \in B_\lambda \mathcal{K}_1(\mathcal{D}_*)$. With $C_{T,\lambda} = L_\lambda[1 + \frac{1}{2}(1 + L_\lambda R_*)^2]$, the projection lemma gives $\eta_m^H \leq C_\lambda C_{T,\lambda} r_m^2$, where $r_m = \|u^* - z_m\|_*$. Orthogonality and the source representation give $S_m^* \geq r_m^2/B_\lambda$. Hence a sufficient relative-score condition is

$$(4.13) \quad \Delta_\lambda = (tc_* - t_0 C_\lambda) - (1 + t)C_\lambda^2 C_{T,\lambda} B_{\text{src}} > 0,$$

with $L_\lambda R_* \leq 1/2$. Invariance of the ball and a quantitative entropy bound remain additional requirements.

To make the entropy-to-rate step explicit, fix a restart index m_0 and set $Q_0 = I - P_{m_0}$, $H_0 = Q_0 H$, $f_0 = Q_0 u^*$, and $\mathcal{D}_{\text{rst}} = Q_0 \mathcal{D}_* \setminus \{0\}$. The projected atoms are not renormalized. Suppose $K = \mathcal{K}_1(\mathcal{D}_{\text{rst}})$ is compact and $f_0 \in B_* K$, $B_* > 0$. For weak OGA of strength t_0 in H_0 , let $d_i \in \mathcal{D}_{\text{rst}}$ be the selected atoms and Π_j the orthogonal projection onto their first j spans, with $\Pi_0 = 0$. The bounded-atom entropy argument of [9, Theorem 4.2] gives

$$(4.14) \quad \|f_0 - \Pi_j f_0\|_* \leq \frac{B_* (j! V_j)^{1/j}}{t_0 \sqrt{j}} \varepsilon_{j+1}(K; H_0), \quad V_j = \frac{\pi^{j/2}}{\Gamma(j/2 + 1)}.$$

Indeed, writing $y_i = \|f_0 - \Pi_i f_0\|_*^2$ and $\ell_i^{\text{rst}} = \|(I - \Pi_{i-1})d_i\|_*$, the source and weak selection give

$$y_i \leq y_{i-1} - \frac{t_0^2 y_{i-1}^2}{B_*^2 (\ell_i^{\text{rst}})^2}, \quad y_j \leq \frac{B_*^2}{t_0^2 j} \left(\prod_{i=1}^j \ell_i^{\text{rst}} \right)^{2/j}.$$

The second bound follows by reciprocation and the arithmetic–geometric mean inequality while the errors are positive. The cross-polytope generated by $\{\pm d_1, \dots, \pm d_j\}$ lies in $\Pi_j K$ and has volume $(2^j/j!) \prod_i \ell_i^{\text{rst}}$. A cover of K by 2^j balls yields $\prod_i \ell_i^{\text{rst}} \leq j! V_j \varepsilon_{j+1}(K; H_0)^j$, by taking the infimum over covering radii. This proves (4.14); zero errors persist. Stirling’s formula bounds $(j! V_j)^{1/j} / \sqrt{j}$ uniformly.

THEOREM 4.7 (Conditional Hessian transfer after restart). *Assume (4.7), (4.11), and exact active-space minimization. At each step $m_0 \leq m < m_0 + J$, suppose nonlinear selection has strength at least t and (4.12) holds with the same $t_0 > 0$. Assume the exact minimizers and frozen projections remain in the Hessian ball for $m_0 \leq m \leq m_0 + J$, K is compact, and*

$$f_0 \in B_* \mathcal{K}_1(\mathcal{D}_{\text{rst}}), \quad \varepsilon_j(\mathcal{K}_1(\mathcal{D}_{\text{rst}}); H_0) \leq C_{\text{ent}} j^{-\beta}, \quad 1 \leq j \leq J + 1,$$

with $B_*, C_{\text{ent}} > 0$, $\beta > 0$. Then, for $1 \leq j \leq J$,

$$(4.15) \quad \|z_{m_0+j} - u^*\|_* \leq C B_* C_{\text{ent}} t_0^{-1} j^{-\beta},$$

$$(4.16) \quad \mathcal{E}(G_{m_0+j}^\sharp) - \mathcal{E}(u^*) \leq C_{\text{nl}} j^{-2\beta}.$$

Here C is absolute and C_{nl} depends on $L_*, R_*, B_*, C_{\text{ent}}, t_0$. An asymptotic claim requires the same constants and hypotheses for all J .

Proof. The relative score proposition gives weak frozen selection after projection by Q_0 , so the restarted spaces form weak OGA on H_0 . Apply (4.14). The projection lemma gives $\|G_{m_0+j}^\sharp - u^*\|_* \leq (1 + L_* R_*) \|z_{m_0+j} - u^*\|_*$; the Taylor estimate then yields the energy bound. $\square$

For the concrete semilinear family, a conditional nonlinear rate therefore requires the invariant-ball condition, the positive margin Δ_λ , and the projected-dictionary entropy estimate simultaneously. The finite-pool numerical trajectory does not verify these conditions uniformly.

4.4. Approximate-source and quartic consequences. Let Y be a finite-dimensional real Hilbert space containing u^* , the active spaces in the finite range. Let $a_*(\cdot, \cdot)$ be its frozen inner product, let P_m be the corresponding orthogonal projection onto $\widehat{\Phi}_m$. Assume $\widehat{\Phi}_{m+1} = \widehat{\Phi}_m + \text{span}\{d_{m+1}\}$ with $d_{m+1} \notin \widehat{\Phi}_m$, and set

$$z_m = P_m u^*, \quad e_m = u^* - z_m, \quad r_m = \|e_m\|_*.$$

Let $\mathcal{D}_{\text{fin}} \subset Y$ be a finite symmetric dictionary containing every accepted direction in this range. For the selected direction d_{m+1} define its innovation norm $\ell_m = \|(I - P_m)d_{m+1}\|_* > 0$ and $S_m^{\text{fin}} = \max_{d \in \mathcal{D}_{\text{fin}}} |a_*(e_m, d)|$. Set $\theta_m = |a_*(e_m, d_{m+1})|/S_m^{\text{fin}}$ when $S_m^{\text{fin}} > 0$, and $\theta_m = 0$ otherwise. Assume that the restarted residual has the explicit approximate atomic representation

$$Q_0 u^* = \sum_{d \in \mathcal{D}_{\text{fin}}} a_d Q_0 d + \rho_0, \quad \sum_d |a_d| \leq B_{\text{entry}}, \quad \|\rho_0\|_* \leq \sigma, \quad Q_0 = I - P_{m_0}.$$

For any $0 \leq \underline{\theta}_m \leq \theta_m$, define $W_j = \sum_{m=m_0}^{m_0+j-1} \underline{\theta}_m^2 / \ell_m^2$.

THEOREM 4.8 (Weighted finite-trajectory source bound). *For $x_0 = (r_{m_0}^2 - \sigma^2)_+ > 0$ and $B_{\text{entry}} > 0$,*

$$(4.17) \quad (r_{m_0+j}^2 - \sigma^2)_+ \leq \left(x_0^{-1} + \frac{W_j}{4B_{\text{entry}}^2} \right)^{-1}, \quad 0 \leq j \leq J.$$

If $x_0 = 0$ or $B_{\text{entry}} = 0$, monotonicity gives $r_m \leq \sigma$ on the range; for $W_j = 0$ the estimate is interpreted through projection monotonicity.

Proof. While $x_m = (r_m^2 - \sigma^2)_+ > 0$, Pythagoras gives $r_m^2 - r_{m+1}^2 = |a_*(e_m, d_{m+1})|^2 / \ell_m^2$. The approximate source first gives $S_m^{\text{fin}} \geq (r_m^2 - \sigma^2)_+ / (2B_{\text{entry}})$. Since $|a_*(e_m, d_{m+1})| \geq \underline{\theta}_m S_m^{\text{fin}}$, Pythagoras yields the scalar recurrence $x_{m+1} \leq x_m - \underline{\theta}_m^2 x_m^2 / (4B_{\text{entry}}^2 \ell_m^2)$, $x_m = (r_m^2 - \sigma^2)_+$. Applying the reciprocal inequality proves the bound until the first index at which $x_m = 0$; after that index monotonicity gives the stated positive-part estimate. $\square$

For the one-dimensional pure or regularized quartic energy, let

$$X = \{u \in W^{1,4}(0, 1) : \int_0^1 u = 0\}, \quad \mathcal{E}_\epsilon(u) = \int_0^1 \left(\frac{\epsilon^2}{2} |u'|^2 + \frac{1}{4} |u|^4 \right) dx - \ell(u),$$

and assume $u^* \in X$ solves its weak equation. Let $Y \subset X$ contain these finite-dimensional spaces and define

$$a_*(v, w) = \int_0^1 (\epsilon^2 + 3|(u^*)'|^2) v' w' dx.$$

For $\epsilon > 0$ this is positive definite on Y ; for $\epsilon = 0$ assume it is positive definite on all of Y . In this paragraph $\hat{\delta}_m = \mathcal{E}_\epsilon(\hat{G}_m) - \mathcal{E}_\epsilon(u^*)$. Then

$$(4.18) \quad \hat{\delta}_m = \frac{1}{2}(r_m^2 + D_m^2) + R_m, \quad R_m = \int_0^1 ((u^*)' v_m^3 + \frac{1}{4} v_m^4) \, dx,$$

where $D_m = \|\hat{G}_m - z_m\|_*$ and $v_m = \hat{G}'_m - (u^*)'$. Thus

PROPOSITION 4.9 (Quartic energy comparison). *If $q_{\text{corr}} \geq 0$, $\rho \geq 0$, $D_m \leq q_{\text{corr}} r_m$, and $|R_m| \leq \rho(r_m^2 + D_m^2)$ with $\rho < 1/2$, then*

$$(4.19) \quad (\tfrac{1}{2} - \rho)r_m^2 \leq \hat{\delta}_m \leq (\tfrac{1}{2} + \rho)(1 + q_{\text{corr}}^2)r_m^2.$$

Proof. Expand the quartic density about $(u^*)'$, cancel the linear term with the weak equation, and use orthogonality of the frozen projection to obtain (4.18); the comparison follows immediately. $\square$

THEOREM 4.10 (Power-type finite-range bound). *Define the normalized frozen innovation decrease*

$$\gamma_m = \frac{|a_*(e_m, d_{m+1})|}{\ell_m}, \quad r_{m+1}^2 = r_m^2 - \gamma_m^2.$$

Let $\alpha > 0$, $c_0 > 0$, and suppose $r_{m_0} > 0$. If $\gamma_m^2 \geq c_0 r_m^{2+2/\alpha}$ on $m_0 \leq m < m_0 + J$, then

$$(4.20) \quad r_{m_0+j}^2 \leq \left(r_{m_0}^{-2/\alpha} + \frac{c_0 j}{\alpha} \right)^{-\alpha}, \quad 0 \leq j \leq J.$$

For the quartic models, Proposition 4.9 converts this into the corresponding energy, natural-distance, and Sobolev bounds on the same finite range.

Proof. Apply convexity to $x^{-1/\alpha}$ with $x_m = r_m^2$ and sum the increments; the quartic comparison then gives the energy and metric statements. $\square$

5. CGA numerical experiments and finite-range estimates.

5.1. Numerical setup, representative problems, and error quantities.

All problems are posed on $\Omega = (0, 1)^d$, with $d = 1$ or 2 , and use manufactured profiles. The main suite uses smooth profiles; the final row is explicitly a low-order activation case $(p, k) = (5, 1)$, so its regularity statement is kept separate. The exact profiles, loads, homogeneous natural boundary data, feasible spaces, and case parameters are specified in Supplement S1. The exact solution is used to construct the load and to evaluate errors, but not in the greedy selection or coefficient optimization. Pure and regularized p -problems use a zero-mean representative, and their atoms are centered before normalization.

For a raw atom φ , let $P_\circ \varphi$ denote centering in the pure and regularized cases and the identity in the other cases. With Q_{tr} denoting the training quadrature, the experimental atom is $g_Q = P_\circ \varphi / \mathcal{N}_Q(P_\circ \varphi)$, where

$$\mathcal{N}_Q(g) = \begin{cases} [Q_{\text{tr}}(|g|^2 + |\nabla g|^2)]^{1/2}, & \text{linear and semilinear cases,} \\ [Q_{\text{tr}}(\sum_i |\partial_i g|^p)]^{1/p}, & \text{pure and regularized } p\text{-cases,} \\ [Q_{\text{tr}}(|g|^p + \sum_i |\partial_i g|^p)]^{1/p}, & \text{reaction } p\text{-case.} \end{cases}$$

This discrete homogeneous normalization is used for atom construction; it is distinct from the theoretically normalized continuous dictionary in Section 2 and from the nonlinear V -distance used to evaluate an approximation. Consequently, a discrete coefficient norm or training-quadrature normalization is not by itself an entropy or source bound in the analytical norm. The componentwise gradient convention agrees with the reported seminorm below; the PDE energy and the vector-valued V -map retain the Euclidean gradient convention.

For coefficient optimization, residuals are evaluated in the stable coordinates supplied by the active feature basis. If a Euclidean coefficient gradient is used, the active Gram matrix is required to convert it to the dual norm on the active span; when the gradient is evaluated with the training quadrature, the corresponding quadrature error must also be included. Thus, an optimizer residual is not by itself a value of η_m^{stat} or $\varepsilon_m^{\text{opt}}$ in the finite-pool descent result Proposition 4.2. The supplementary numerical data report the projected residual, the common target, the coefficient-solve convergence, and the energy decrease as separate quantities.

At each trial, the available atom with largest derivative score is tested, and all active outer coefficients are reoptimized when the innovation test succeeds. A retained step m is one that passes the rank, finite-value, stationarity, energy-decrease, and feasibility tests. Its active rank is N_m , while M denotes the terminal retained step; rejected directions do not change m or N_m . The retained atoms are linearly independent, so the plotted width is $N = N_m = m$. Selection budgets and stopping criteria are specified in Supplement S1. The eight cases in Table 1 include a quadratic calibration, two semilinear energies, three $p = 4$ diffusion structures, a one-dimensional pure $p = 4$ problem, and a low-regularity $p = 5$ case.

TABLE 1
Representative CGA experiments and terminal retained widths.

Energy	d	p	k	Retained/target
linear reaction–diffusion, multifrequency	1	2	3	256/256
cubic semilinear, multifrequency	1	–	3	256/256
hyperbolic-sine semilinear	2	–	3	512/512
pure p -Laplacian, low-frequency profile	1	4	3	141/256
pure p -Laplacian	2	4	3	512/512
regularized p -Laplacian, $\varepsilon = 0.1$	2	4	3	512/512
reaction p -Laplacian	2	4	3	512/512
pure p -Laplacian	1	5	1	256/256

For the main eight-case suite, the one-dimensional target width is 256 and the two-dimensional target width is 512. The separate quartic auxiliary numerical tests use $N = 8$ –32 for calibration and continue through $N = 64$. The one-dimensional pure $p = 4$ case reaches 141 retained atoms within 768 attempted selections and therefore has $N = 128$ as its last dyadic state. The CGA curves show the retained states over the reported range. Markers identify the powers of two used for local-order tables and are the only states joined by the display curves; the values at all retained widths are supplied with the supplementary numerical data.

The quartic finite-trajectory calculation uses two finite index ranges. The constants and source quantities are calibrated only on the 25 retained states $N = 8, \dots, 32$; the states $N = 33, \dots, 64$ continue the same trajectory with these constants held fixed. Consequently, the number of transitions satisfying the normalized-innovation condition

on $32 < N \leq 64$ measures whether the calibration remains informative immediately beyond its calibration range, whereas only the $8 \leq N \leq 32$ quantities enter the stated finite-range envelope. The two transition sets are reported separately below.

For the linear and semilinear cases we report the energy gap and

$$e_{H^1}(\widehat{G}_N) = \frac{\|\widehat{G}_N - u^*\|_{H^1}}{\|u^*\|_{H^1}}.$$

For the pure and regularized p -models we report the relative gradient seminorm, with the componentwise convention

$$|v|_{1,p} = \left(\int_{\Omega} \sum_{i=1}^d |\partial_i v|^p dx \right)^{1/p}, \quad e_{\nabla,p}(\widehat{G}_N) = \frac{|\widehat{G}_N - u^*|_{1,p}}{|u^*|_{1,p}}.$$

For the reaction p -model, the reported relative error uses the full norm $\|v\|_{W^{1,p}} = (\|v\|_{L^p}^p + |v|_{1,p}^p)^{1/p}$. On the zero-mean space the gradient seminorm and full norm are equivalent, so the theoretical error powers apply to both, but their relative numerical values are not identified. The relative natural distance is $e_V(\widehat{G}_N) = d_{V,\bullet}(\widehat{G}_N, u^*)/d_{V,\bullet}(0, u^*)$. The benchmark denominators are nonzero. Energy differences retain their sign: $\widehat{\delta}_N = \mathcal{E}(\widehat{G}_N) - \mathcal{E}(u^*)$; unresolved negative quadrature values are excluded from energy summaries rather than replaced by absolute values. The dyadic local order is

$$\text{ord}_N(e) = \log_2 \left(\frac{e(N/2)}{e(N)} \right).$$

These three quantities are kept separate in every figure, table, and subsequent comparison.

5.2. Conditional reference rates on finite index ranges. For ReLU^k atoms, the Hilbert-space approximation exponent associated with the smooth-ridge entropy estimate is

$$\beta_H = \frac{1}{2} + \frac{2(k-1)+1}{2d}.$$

The dashed segments in the CGA convergence figures show the conditional powers $N^{-2\beta_H}$ for the energy gap and $N^{-\beta_H}$ for either the H^1 error or the natural V -distance. For a standard p -structure energy, the corresponding comparison power for the $W^{1,p}$ error is $N^{-2\beta_H/p}$, as follows from the energy-to-metric conversion in Corollary 3.7. The exponent is fixed by the stated theory and is not fitted to the curve. The vertical position is chosen once from the last up to three positive dyadic states by a median in log scale. The displayed segments are conditional reference curves, not fitted exponents and not evidence that the continuous coverage, source, and transfer assumptions hold on the plotted range.

5.3. Linear and cubic calibration. Figure 1 shows the retained dyadic states for the linear and cubic multifrequency problems. Both curves decrease rapidly after about 32 atoms. Between $N = 128$ and 256, the linear energy and H^1 orders are 6.083 and 3.042, whereas the cubic orders are 6.896 and 3.448; the corresponding terminal relative H^1 errors are 1.247×10^{-4} and 1.215×10^{-4} . These are measured local orders on one resolved interval, not estimates of an asymptotic exponent.

The two-dimensional sinh case is reported in the supplementary figures. Its relative H^1 error decreases from 4.074×10^{-3} at $N = 256$ to 1.990×10^{-3} at $N = 512$, while

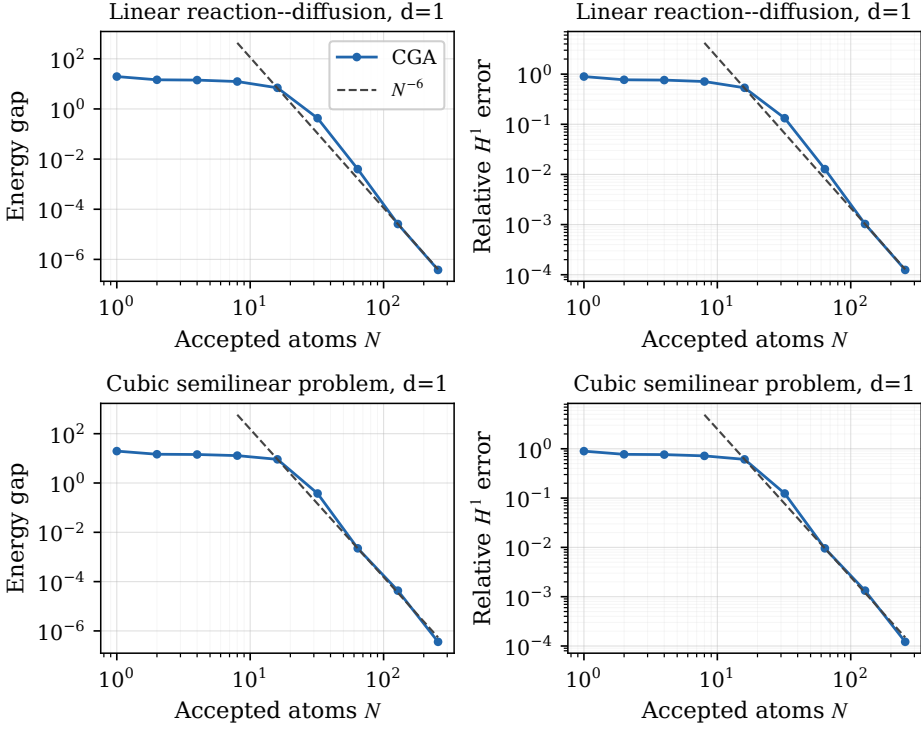


FIG. 1. CGA values at retained powers of two for the linear and cubic problems. Each dashed segment is the corresponding metric-specific conditional reference rate of the displayed conditional exponent, restricted to the displayed index range.

the local order decreases from 1.793 on $128 \rightarrow 256$ to 1.034 on $256 \rightarrow 512$. The final interval therefore exhibits slower decay than the preceding interval.

5.4. Numerical results for the pure and regularized $p = 4$ models. Figure 2 presents the one- and two-dimensional pure $p = 4$ cases using energy, relative gradient L^4 seminorm error, and relative V -distance. The one-dimensional case reaches relative errors 1.579×10^{-5} in the gradient seminorm and 1.680×10^{-6} in the V -distance at its last dyadic state $N = 128$. For the two-dimensional pure $p = 4$ problem, the interval $128 \rightarrow 256$ has energy and V -orders 3.410 and 1.703. On the final interval the V -distance continues to decrease, but the relative gradient seminorm error increases from 2.494×10^{-3} to 3.601×10^{-3} . Thus the terminal Sobolev-error behavior is nonmonotone even though the V -distance decreases.

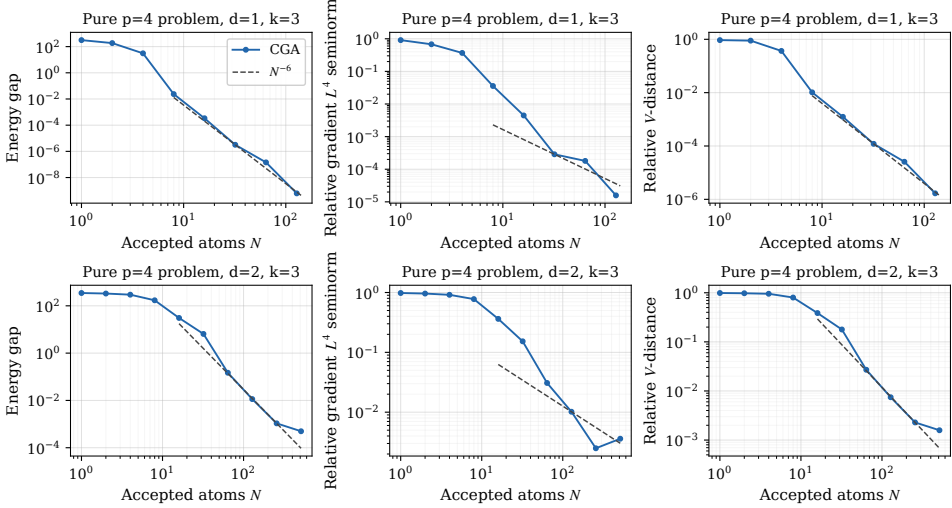


FIG. 2. CGA values at retained powers of two for the representative one- and two-dimensional pure $p = 4$ problems. The three columns show energy gap, relative gradient L^4 seminorm error, and relative V -distance. Dashed segments use the metric-specific conditional exponents stated above.

The dyadic values for the semilinear and $p = 4$ cases are supplied in the supplementary numerical records. The regularized and reaction trajectories show the same separation between the natural-distance and Sobolev metrics as the representative pure cases.

5.5. Finite-trajectory checks for the quartic models. We evaluate the projection and quartic estimates of Section 4 on the one-dimensional pure and regularized $p = 4$, $k = 3$ trajectories with $u^*(x) = \cos(2\pi x)$ and $\varepsilon = 0$ or 0.1. The constants are calibrated on the retained states $N = 8, \dots, 32$ and then held fixed on the continuation to $N = 64$. The calibration and continuation ranges are kept separate, so the latter is only a finite-range check.

On the calibration range, the quartic identity and the projection-drop identity agree with the computed values to floating-point accuracy under two quadrature orders. The source and quartic comparison bounds cover the computed energy errors in both models, although their terminal constants are conservative. For the prescribed benchmark $\alpha = 6$, the resulting finite-range comparison powers are

$$\begin{aligned} \widehat{\delta}_{8+j} &= O(j^{-6}), & d_{V,\bullet}(\widehat{G}_{8+j}, u^*) &= O(j^{-3}), \\ \|\widehat{G}_{8+j} - u^*\|_{W^{1,4}} &= O(j^{-3/2}), & & 1 \leq j \leq 24. \end{aligned}$$

These are conditional envelopes on the calibrated range and do not assert an asymptotic rate. The normalized-innovation condition remains satisfied for 27 of 32 continuation transitions in the pure model and 28 of 32 in the regularized model; the exceptions prevent extending the same fixed-constant bound to the whole continuation.

The detailed per-state source quantities, Hessian margins, quadrature checks, and continuation records are supplied in the supplementary numerical files.

6. Equal-DOF baseline comparisons.

6.1. Comparison setup and scope. Five cases use a common evaluator: linear and cubic in one dimension, sinh in two dimensions, and pure $p = 4$ in one and two dimensions. Counts are retained CGA/RFM coefficients and independent FEM coefficients after removing the constant mode where required; equal counts imply neither equal numerical rank nor equal computational work.

The `review_replay_20261001` batch refits coefficients on fixed archived CGA greedy prefixes, separately from the original training trajectories in Section 5. FEM uses uniform one-dimensional or structured triangular meshes with P1/P3 references and supplementary P2 states. RFM uses nested normalized ReLU³ ridges and ten prespecified seeds, 201–210, per width. The archive has 320 RFM, 32 CGA-prefix, and 114 FEM states, including failures. Sampling, fitting, quadrature, and audit settings are frozen in the protocol specified in S4. Historical baseline statistics are retained separately.

Figures join computed dyadic states and terminal CGA prefixes; FEM uses its mesh counts. Display caps are the archived CGA terminal widths. Table endpoints are fixed at 256, 256, 256, 128, 256 for C1–C5. FEM tables interpolate $\log(\text{error})$ against $\log(\text{DOF})$ between metric-valid states, without extrapolation; P3 meshes do not bracket 512 in two dimensions. These count-based choices precede inspection of the reported errors. Figures contain neither common-grid interpolants nor theoretical reference lines.

6.2. Evaluation and statistical samples. Every method uses the same evaluator and metric-specific rule: a successful solve, a finite nonnegative metric, and successive permitted evaluation-quadrature relative difference at most one percent. Signed negative energy gaps remain raw and do not enter energy summaries; an energy exclusion does not itself exclude a stable Sobolev or V observation. Quadrature agreement is a numerical diagnostic, not a rigorous enclosure.

RFM medians and interquartile ranges use each metric’s valid observations. The denominator remains ten; *Valid* in Table 2 counts its Sobolev sample. Incomplete samples are conditional comparisons. S3 supplies solve/audit reasons, sample counts, and model hashes.

6.3. Relative Sobolev errors. C1–C3 report relative H^1 errors. C4–C5 report the relative componentwise gradient L^4 seminorm $e_{\nabla,4}$ from Section 5, rather than the relative full $W^{1,4}$ norm. The natural V -distance is separate.

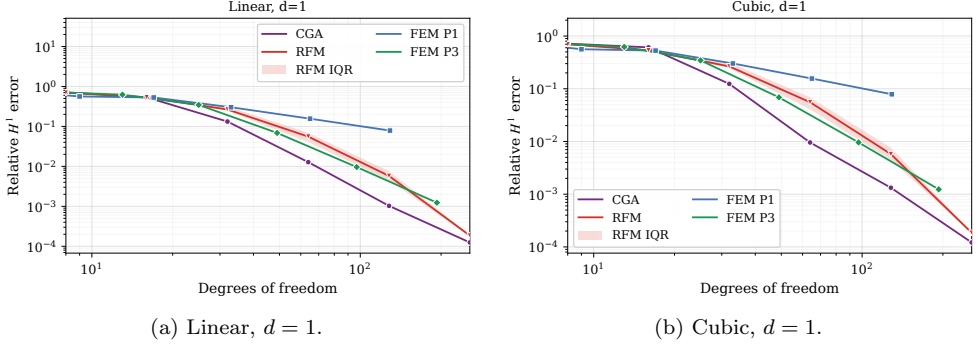


FIG. 3. Relative H^1 error versus coefficient count for the one-dimensional comparison batch. RFM bands show metric-valid medians and interquartile ranges from ten prescribed seeds. Markers are computed states; CGA coefficients are refitted on fixed greedy prefixes.

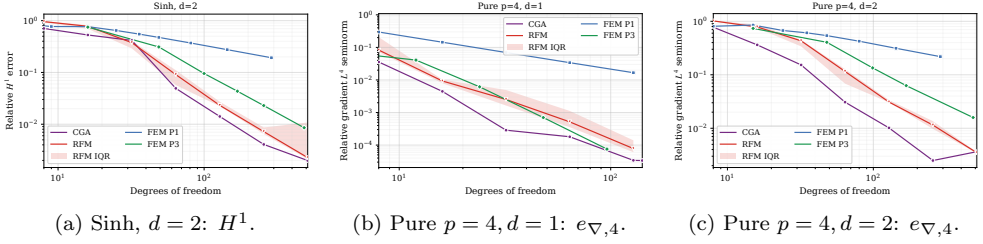


FIG. 4. Relative H^1 error for $\sinh$ and relative gradient L^4 seminorm for the pure $p = 4$ problems. RFM bands use observations passing that metric's solve and quadrature checks. The prescribed denominator is ten; endpoint valid counts appear in Table 2.

Table 2 reports endpoint values and ratios. A ratio above one favors CGA at that count; a ratio below one favors the comparison method. Rankings depend on the problem, count, evaluation policy, and valid sample. These data establish neither uniform dominance at every DOF nor a ranking over excluded realizations. For the one-dimensional pure $p = 4$ case at $N = 128$, the P3/CGA ratio is about 0.88 and the RFM-median/CGA ratio is about 2.30. Thus the interpolated P3 reference has a slightly smaller error than CGA at this endpoint, while CGA is more accurate than the specified RFM median.

TABLE 2

Relative H^1 error for C1–C3 and relative gradient L^4 seminorm for C4–C5 at the prescribed common coefficient counts. RFM statistics are median [Q1,Q3] over metric-valid realizations from ten prescribed seeds. Valid means a successful solve, a finite nonnegative metric, and an at-most-one-percent successive-quadrature difference for that metric; incomplete groups are conditional comparisons. Ratios larger than one favor CGA.

Problem	DOF	CGA		RFM median [Q1,Q3]		Valid	RFM/CGA	FEM P3/CGA
Linear, $d=1$	256	1.247×10^{-4}	1.853×10^{-4}	$[1.779 \times 10^{-4}, 1.913 \times 10^{-4}]$		10/10	1.49	4.24
Cubic, $d=1$	256	1.215×10^{-4}	1.853×10^{-4}	$[1.779 \times 10^{-4}, 1.913 \times 10^{-4}]$		10/10	1.52	4.35
Sinh, $d=2$	256	4.074×10^{-3}	7.196×10^{-3}	$[6.472 \times 10^{-3}, 8.663 \times 10^{-3}]$		10/10	1.77	5.63
Pure $p=4$, $d=1$	128	3.395×10^{-5}	7.818×10^{-5}	$[6.046 \times 10^{-5}, 1.400 \times 10^{-4}]$		10/10	2.30	0.88
Pure $p=4$, $d=2$	256	2.493×10^{-3}	1.147×10^{-2}	$[9.730 \times 10^{-3}, 1.360 \times 10^{-2}]$		6/10	4.60	14.54

6.4. Natural V -distance for the p -models. Figure 5 uses the relative V -distance from Section 5. It is separate from the gradient seminorm: the metrics need not have the same terminal behavior or valid sample. Computed-state display and metric-specific RFM uncertainty conventions are otherwise the same.

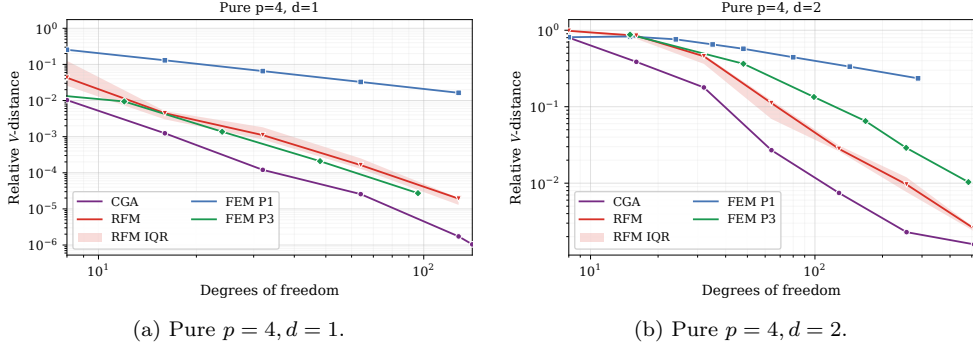


FIG. 5. Relative V -distance versus coefficient count for the comparison batch, with metric-valid RFM medians and interquartile bands. No theoretical reference line or common-grid interpolation is drawn.

6.5. Interpretation limits. These comparisons cover the specified archived fits, without establishing runtime ordering, asymptotic complexity, or rankings over excluded realizations. Energy gaps may remain unresolved while another metric is stable. S3–S4 give solve statuses, audit exclusions, interpolation brackets, and coefficient models supporting the figures and tables.

7. Discussion and conclusion. This work develops a unified variational framework for the analysis of finite-pool Chebyshev greedy algorithms (CGA) for nonlinear PDEs. The atomic source condition and directional smoothness yield an $O(m^{-1})$ energy bound for the ideal continuous-dictionary CGA. For the practical finite-pool algorithm, the analysis further separates the effects of continuous-dictionary coverage, score-evaluation error, radial stationarity, and inexact coefficient optimization, and identifies relative accuracy conditions under which the same energy order is retained. This provides a quantitative description of how finite-pool approximation, numerical score evaluation, and inexact active-space solves influence the greedy descent mechanism.

For the standard pure, regularized, and reaction p -energies with $p \geq 2$, we further establish quantitative relations among the energy gap, the natural V -distance, and the Sobolev error. In particular, an energy exponent α leads to the corresponding natural-distance and $W^{1,p}$ -error exponents $\alpha/2$ and α/p , respectively. This allows energy estimates for the greedy algorithm to be translated directly into error measures adapted to the nonlinear geometry of the underlying PDEs.

Building on this structure, two refined analytical mechanisms are developed. The local Hessian-transfer analysis connects the nonlinear problem with greedy approximation in a frozen Hilbert metric, while the finite-trajectory analysis works directly with the nested spaces generated by the computation and produces explicit projection bounds through either an approximate source representation or normalized innovations. For the quartic energy, an exact energy decomposition further connects these projection estimates with nonlinear energy errors. Consequently, even on computational

ranges where a uniform asymptotic theory is not yet available, the analysis provides quantitative and directly testable explanations of the observed algorithmic trajectories.

The one-dimensional pure and regularized $p = 4$ experiments support this finite-trajectory framework. On the calibration range $N = 8\text{--}32$, both the source-based and fractional-innovation bounds enclose the computed errors. For the prescribed benchmark $\alpha = 6$, the conditional comparison powers on this range are 6, 3, and $3/2$ for the energy, natural distance, and $W^{1,4}$ error, respectively. The projection and quartic identities remain accurate under two quadrature rules on the continuation to $N = 64$. The normalized-innovation sufficient condition holds for 27 of 32 continuation transitions in the pure model and 28 of 32 in the regularized model. Numerical coverage by a conservative expression on the continuation does not establish the same theorem guarantee throughout that range.

The equal-coefficient comparisons use refitted CGA prefixes and fully archived FEM and RFM states with the same metric-specific quadrature policy. Their computed CGA and RFM endpoints, together with interpolated FEM reference values where indicated, describe the approximation errors for the specified problems and valid samples. They do not establish uniform superiority over all widths, excluded realizations, or computational-work budgets. The componentwise gradient seminorm, full Sobolev norm, and natural distance are reported as distinct metrics rather than interchangeable relative errors.

Overall, the theoretical and numerical results show that finite-pool CGA admits a coherent variational analysis while providing effective low-dimensional adaptive approximations for nonlinear PDEs with p -structure. The present framework also identifies concrete and testable conditions for further developments, including continuous-dictionary coverage, entropy bounds for the restarted dictionary, and invariance of the Hessian neighborhood. Establishing these conditions uniformly, together with reliable numerical enclosures and comparisons under a common computational-work measure, would provide a natural route from the present finite-range theory toward a more complete asymptotic convergence and complexity theory.

8. Code and data availability. Code and data are publicly available at <https://github.com/RonzolYu/cga4PDE>. The comparisons use the `review_replay_20261001` batch and `code/config/review_replay.json`. Each comparison state has a coefficient model/hash, evaluation rule, and metric-specific audit. Historical records remain separate. The README and S3–S4 specify sample definitions and distinguish artifact rebuilding, saved-model reevaluation, and fresh coefficient fits.

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